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Similarity in L2 Phonology: Evidence from L1 Spanish late-learners' perception and lexical representation of English vowel contrasts

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Abstract

Adult second language (L2) learners often experience difficulty producing and perceiving nonnative phonological contrasts. Even relatively advanced learners, who have been exposed to an L2 for long periods of time, struggle with difficult contrasts, such as /ɹ/–/l/ for Japanese learners of English. To account for the relative ease or difficulty with which L2 learners perceive and acquire nonnative contrasts, theories of L2 speech perception and phonology often appeal to notions of 'similarity', but how is 'similarity' best captured? In this article, we review two prominent approaches to similarity in L2 speech perception and phonology and present the findings from two experiments that investigated the role of phonological features in the perception and lexical representation of two vowel contrasts that exist in English, but not in Spanish. In particular, we explored whether L1 phonological features can be reused to represent nonnative contrasts in the second language (Brown, 1998, 2000), as well as to what extent new phonological structure might be acquired by advanced late-learners. We show that second language acquisition of phonology is not constrained by the phonological features made available by the learner's native language grammar, nor is the use of particular phonological features in the native language grammar sufficient to trigger redeployment. These findings suggest that feature availability is neither a necessary, nor a sufficient, condition to predict the observed learning outcomes. These results are discussed in the context of current theories of nonnative and L2 speech perception and phonological development.

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I Introduction

It is well established that patterns of cross-language and nonnative speech perception and production are shaped by the sound structure of the listeners' native language. Adult second language learners typically have accents that are identifiable because they differ from target language norms in systematic ways that are characteristic of speakers who share language background. Even after years of exposure to the target language, second language learners may struggle with the perception and production of 'difficult' contrasts, such as /ɹ/–/l/ for Japanese learners of English (Goto, 1971; Logan et al., 1991; Miyawaki et al., 1975; Sheldon and Strange, 1982; Yamada, 1995), or /i/–/ɪ/ for Spanish learners of English (Escudero and Boersma, 2004; Flege et al., 1997; Fox et al., 1995). Models of nonnative and second language (L2) speech perception and phonological development typically appeal to notions of 'similarity' in order to account for the relative ease or difficulty of the perception and production of various nonnative and target language sound contrasts (Best, 1995; Brown, 1998, 2000; Escudero, 2005, 2006, 2009; Flege, 1995). But, how is similarity in L2 phonology best captured to account for observed patterns of perception and production? What are the representations that guide L2 sound learning? And, to what extent can the influence of the native language sound system be overcome?

Traditionally, work in L2 speech perception and phonology has reflected two broad approaches to similarity in the cognitive sciences, namely spatial approaches (Shepard, 1962, 1987) and featural approaches (Tversky, 1977). These approaches make different representational commitments and, therefore, provide different answers to the questions posed above. We discuss these two approaches in turn.

II Review

1 Spatial approaches to similarity in L2 phonology

In spatial models of similarity (Shepard, 1962, 1987), objects are represented as points in a continuous, multidimensional space, and similarity between objects corresponds to metric distance between the corresponding points. The closer together the two objects are in this representational space, the more similar they are; the further apart, the more dissimilar. In spatial models of cross-language speech perception, the relevant space is perceptual, articulatory, or acoustic. Given that phonetic descriptions of the physical implementation of speech sounds in production and the interpretation of fine-grained acoustic cues in perception lend themselves well to these types of spatial comparisons, spatial models of similarity have been adopted by those who favor phonetic explanations of L2 sound learning.

Perceptual similarity is at the core of Flege's Speech Learning Model (SLM) (Flege, 1995, 2003), which aims to account for the ease and difficulty with which nonnative sounds are produced and perceived by second language learners. The learners' task is viewed as one of determining where new target language sounds lie in a multidimensional space. The predictions of the model are based on the proximity of L2 sounds to first language (L1) phonetic categories. The model holds that, although adult L2 learners maintain the capacity to acquire new categories, their success can be predicted by whether or not a target language phone is 'identical', 'new', or 'similar'. Phones that are 'identical' in both the L1 and L2 are not expected to present difficulty due to the process of 'positive transfer' (Weinreich, 1953). 'New' phones are those that are different from any L1 category and are, therefore, perceived with little interference from the native language. 'Similar' phones, in contrast, are predicted to pose difficulties as a result of the process of 'equivalence classification' whereby L2 sounds are equated to native language categories. From this perspective, in order for a learner to acquire new phonetic categories, he or she must discern the differences between target language sounds and existing phonetic categories. L2 sound learning is expected to be slower and more difficult in the case of 'similar' phones.

Best's Perceptual Assimilation Model (PAM) (Best, 1994, 1995) and Best and Tyler's extension of the model for L2 learners (PAM-L2) (Best and Tyler, 2007) rely on phonetic similarity in order to make predictions about the relative ease and difficulty that nonnative and L2 listeners will experience discriminating novel sound contrasts. The central premise of Best's PAM, which is based in Articulatory Phonology (Goldstein and Fowler, 2003), is that nonnative sounds are mapped onto a listener's native categories on the basis of articulatory similarities. According to the model, the discriminability of a given nonnative contrast is determined by the manner and degree to which the sounds can be assimilated to a listener's native categories. The greatest perceptual difficulty is expected when two different nonnative sounds are mapped to a single L1 category (i.e. Single-Category Assimilation). On the other hand, nonnative sounds that are not assimilated to any L1 sounds (i.e. Unassimilated) or that map to two distinct L1 categories (i.e. Two-Category Assimilation) will be discriminated well, regardless of experience with the language.

While both Flege's SLM and Best's PAM and PAM-L2 are intuitively appealing, and a large body of experimental results have been interpreted within both frameworks (Best et al., 1988, 2001; Best and Strange, 1992; Flege, 1987; Hallé et al., 1999), the proposals suffer from a significant limitation. The models rely heavily on the notion of perceptual similarity to make predictions about second language learners' ability to perceive and acquire various L2 sounds, but neither adequately defines or operationalizes the construct. As a result, there is no a priori means of determining the similarity between L1 and L2 sounds independently of nonnative perceptual patterns that they aim to explain.

Flege (1992) discussed one possible means of categorizing nonnative phones as 'identical', 'similar', and 'new'. The approach takes into account (1) the IPA symbol used to represent the L1 and L2 phones, (2) the acoustic difference between the two, and (3) the perceived difference between the two. According to these criteria, 'similar' phones are those that are represented by the same IPA symbol in both the L1 and L2, but nevertheless differ acoustically from the nearest L1 sound. 'New' sounds then would be acoustically different phones that also are represented by different IPA symbols in the L1 and L2. Yet, Flege himself discusses problems with this approach. Instead, researchers working in this framework seem to rely implicitly on some measure of the acoustic similarity between L1 and L2 phones.

Other cross-language speech-perception researchers have argued explicitly that acoustic similarity between L1 and L2 sounds can predict the perception behavior of nonnative listeners. In her Second Language Linguistic Perception (L2LP) model, Escudero (2005, 2009) claims a direct link between the perception and production of sounds, such that native listeners' perceptual behavior closely matches the relative use of the acoustic cues in the production of their L1 variety (Escudero and Boersma, 2004). L1 perceptual weightings of acoustic cues are adopted in the initial stage of L2 acquisition. In the L2LP model, the initial stage of L2 perception is hypothesized to involve the perceptual mapping of L2 sounds onto L1 categories. Moreover, describing how listeners of one language categorize the sounds of another language provides a measure of the L2 initial state. Escudero and colleagues have produced a growing body of research demonstrating that the acoustic similarity between L1 and L2 sounds influences nonnative perception (Escudero and Chládková, 2010; Escudero and Vasiliev, 2011; Escudero and Williams, 2011; Escudero et al., 2012) and makes predictions regarding the initial state of L2 acquisition and the degree of difficulty that an L2 learner will experience during L2 development. Specifically, L2 contrasts that assimilate to separate L1 categories ('similar scenario' in L2LP) are predicted to be easier to learn than L2 contrasts that assimilate strongly to a single L1 category ('new scenario' in L2LP).

These findings are in contrast with a number of previous studies that suggest that perceptual similarity is not always straightforwardly predicted by acoustic similarity alone. For example, Polka and Bohn (1996) report that L1 Canadian English speakers perceive German /u/ to be a better exemplar of English /u/, than German /y/, despite the fact that German /y/ and English /u/ are acoustically closer. Similarly, Strange et al. (2004) found that L1 American English listeners equate German /y/, /œ/ and French vowels /y/, /ʏ/, /œ/, /ø/ with English back vowels, despite them being acoustically closer in the F2 dimension to the front unrounded vowels of English. Moreover, as discussed by Best and Tyler (2007), English learners of French and French learners of English map French /ʁ/ to English /ɹ/, despite the fact that these sounds are acoustically quite different.

Acoustic similarity does not appear to make the correct predictions with respect to native Spanish listeners' production and perception of English /i/ and /ɪ/ either. Figure 1 shows the mean F1 and F2 of the five Spanish vowels (/i/, /e/, /a/, /o/, and /u/) and two English vowels (/i/ and /ɪ/), plotted in F2 × F1 space.[1] The means of the Spanish categories (Quilis and Esgueva, 1983) are shown as black dots and the means of the two English vowel categories (Hillenbrand et al., 1995) as gray squares. What can be readily observed from the plot is that the mean of the English vowel /i/ lies between the Spanish vowels /i/ and /e/, although a bit closer to the Spanish vowel /i/, whereas the English vowel /ɪ/ lies remarkably close to the Spanish vowel /e/. If acoustic similarity were a good predictor of perceptual similarity, we would expect that Spanish listeners would map English /i/ to Spanish /i/ and English /ɪ/ to Spanish /e/, as these are acoustically the nearest L1 categories. However, anecdotal and empirical observations from both vowel substitutions in production (Amastae, 1978; Brennan and Brennan, 1981; Hammond, 1986; MacDonald, 1989) and interlingual vowel identifications in perception (Flege, 1991;

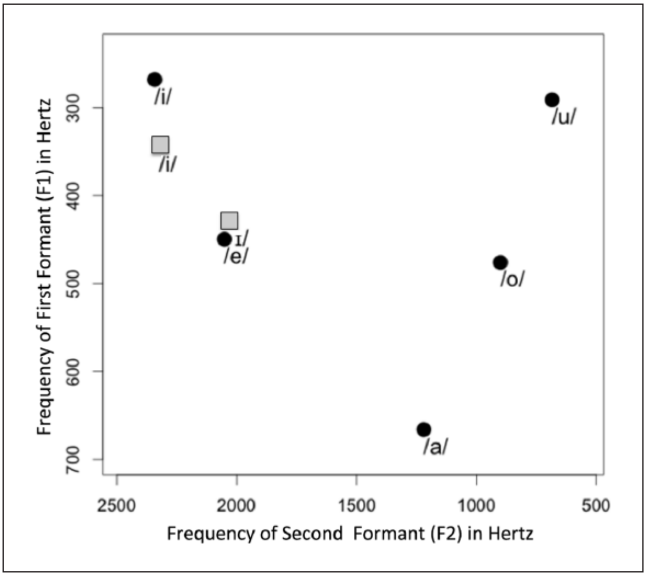

Figure 1. Five Spanish monophthongs (black dots) and two English vowel categories (gray squares) plotted in F2 by F1 space.
Source. The English formant values are from Hillenbrand et al. (1995); the Spanish data are from Quilis and Esgueva (1983).

Scholes, 1967) indicate that Spanish speakers tend to map both English /ɪ/ and English /i/ onto Spanish /i/, not Spanish /e/.

Given the limitations of these spatial approaches to similarity in L2 phonology at present to make a priori predictions about patterns of production and perception – and additional evidence that approaches that are based solely on acoustic similarity are at best incomplete – we investigate whether a feature-based alternative to L1 interference can better account for observed patterns of L2 perception and representation.

2 *Featural approaches to similarity in L2 phonology*

Feature-based models of similarity comparison provide another perspective on the nature of the phonological representations that guide L2 sound learning. In feature-based models of similarity, objects are represented as sets of discrete features, and the similarity of a pair of objects can be determined by comparing the relative size of the sets of common and distinctive features (Tversky, 1977).

Such feature-based representations of mental objects have been advocated for in generative phonology where distinctive features serve as primitives of phonological computation (processes and alternations) and lexical representation (contrast). Indeed, the standard view from linguistics holds that words are encoded in long-term memory in terms of discrete, abstract sequences of phonemes. Phonemes themselves are composed of distinctive features that are organized into hierarchical constituents (Clements, 1985;

Dresher, 2009). These phonological representations are economical in the sense that they contain only unpredictable information necessary to uphold contrast. Predictable aspects of pronunciation are supplied prior to phonetic implementation. Models of speech perception have also adopted abstract, minimally specified lexical representations as a working assumption, permitting the recognition of phonological categories in the face of variation introduced by phonological context, speaker, speech rate, etc. (Lahiri and Marslen-Wilson, 1991; Lahiri and Reetz, 2002, 2010; Poeppel et al., 2008; Stevens, 2002). From a phonological perspective then, the task of the L2 learner with respect to novel target-language contrasts is to build unique segmental representations for contrastive target-language sounds, so as to represent the contrast in the L2 lexicon and to use this knowledge of the contrast to distinguish L2 words.

Brown's Feature-based Model (FBM) of L1 interference provides a phonological approach to L2 speech perception and phonological acquisition in which distinctive features play a central role as the representational building blocks (Brown, 1998, 2000). The central premise of her proposal is that 'the learner's native grammar constrains which nonnative contrasts he or she will be able to accurately perceive and, therefore, limits which nonnative contrasts the learner will successfully acquire' (Brown, 2000: 19). In particular, Brown hypothesizes that if a learner's grammar lacks the feature necessary to uphold a given phonological contrast, he or she will be unable to accurately perceive and, therefore, acquire the nonnative distinction. Yet, the learner may reuse L1 phonological features in order to build segmental representations that distinguish a novel L2 contrast (whether or not these features are the ones actually employed in the target language). Thus, the presence of the necessary phonological feature in the native grammar may facilitate the perception and acquisition of the nonnative contrast, regardless of whether or not the target language segment can be found in the native inventory. Archibald (2005) later characterized this strategy of reuse of L1 features as 'redeployment', with his Redeployment Hypothesis (for a more detailed discussion, see Archibald, 2005, and subsequent work). We adopt Archibald's terminology here.

The predictions of Brown's FBM have been supported empirically by cross-language speech perception studies involving L2 learners from various language backgrounds. In her own work, Brown investigated whether the perception and acquisition of L2 phonological contrasts was limited by the features made available by the L1 grammar. Brown (1998) used an auditory AX discrimination task and a forced choice picture identification task to compare the performance of Mandarin Chinese and Japanese learners on the English /ɹ/–/l/ contrast. She found that Chinese listeners were able to accurately perceive minimal pairs containing the English /ɹ/–/l/ contrast, whereas Japanese listeners were not. According to Brown, this is because the [coronal] feature that is needed to distinguish the contrast is used for representing segmental contrasts in the L1 grammar of the Mandarin Chinese speakers, but not that of Japanese speakers. In a second experiment, Brown investigated the perception and representation of the nonnative /ɹ/–/l/, /b/–/v/ and /f/–/v/ contrasts by Japanese learners. Importantly, Japanese lacks the labiodentals [f] and [v], though it does have a voiceless bilabial fricative [ɸ] as an allophone of /h/. Since the Japanese grammar uses both [continuant] and [voice] features in its system of contrasts (i.e. for native contrasts /t/–/s/ and /p/–/b/, respectively), but does not employ the feature [coronal], it was predicted that Japanese learners would accurately perceive the

English /b/–/v/ and /f/–/v/ contrasts, but not the /ɹ/–/l/ contrast. The results were consistent with this prediction and were taken to support her feature-based model of L1 interference in which new features cannot be acquired, but new segmental representations can be built from existing L1 features. Additional support for the model has since been provided by Matthews (1997) and Atkey (2001).

In contrast, other empirical findings suggest that L2 learners may be able to acquire new structures using features that are non-contrastive in the learner's native language, but which are relevant for phonetic realization in the L1. For example, Curtin et al. (1998) conducted a perception training study to investigate the role of lexical and surface representations in phonological transfer by comparing native English and French-speaking participants' ability to acquire the Thai three-way stop voicing distinction (e.g. /b/–/p/–/pʰ/). They found that both English speakers and French speakers readily acquired the voicing contrast (presumably by redeployment of their native [voice] feature, which is used to distinguish lexical items in their L1). Additionally, the English-speaking group, but not the French-speaking group, showed evidence of acquiring the ability to lexically encode aspiration in Thai. This difference in outcomes is attributed to the presence of aspiration in surface representations in English, but not French. The authors concluded that predictable, non-contrastive aspects of L1 surface representations may be lexicalized in L2 acquisition, even though they are not transferred initially (cf. Pater, 2003). These findings suggest that L2 acquisition of phonology may not be constrained by the set of distinctive features that operate in the L1 phonology, but that experience with non-contrastive features may also play a facilitatory role in L2 phonology.

LaCharité and Prévost (1999) also investigated whether L2 learners can acquire phonological features that are inactive in their native language. The authors explored the acquisition of the sounds /θ/ and /h/ by Canadian French-speaking learners of English. They find that the French-speaking participants showed better performance for /θ/ than they did for /h/. Since the representation of /θ/ relies on the learner's ability to acquire a new terminal feature [distributed] and the /h/ on adding a place feature [pharyngeal], the authors argue that new terminal features can be acquired, but that learners cannot acquire new place features. (For additional evidence suggesting that new features may be learned, see also González Poot, 2011.)

In sum, a growing body of literature bears on the issues of feature redeployment and the acquisition of new phonological features by adult L2 learners. What is noteworthy however is that, as of yet, all evidence regarding these issues has been limited to consonants. That is, no study that we are aware of has brought data from nonnative vowel perception to bear on these issues of feature redeployment and the acquisition of new structure.

3 Phonetic categorization and lexical encoding of novel contrasts

Finally, there has been a growing interest in investigating the relationship between a listener's phonetic categorization abilities and his or her abilities to lexically encode novel contrasts. Traditionally, it has been assumed that there is a one-to-one mapping between phonetic categorization and lexical contrast. That is, if a listener cannot reliably distinguish a pair of phones then he or she will fail to lexically encode a novel contrast and will represent a minimally contrastive pair of words that are distinguished by that contrast as homophones in his or her interlanguage lexicon. The results of Pallier et al. (2001) are compatible with this view. The authors report spurious priming during auditory lexical access by Spanish-dominant Spanish–Catalan bilinguals for minimal pairs containing the Catalan-specific contrasts (i.e. /netə/–/nɛtə/). The authors take this to suggest that, not only do Spanish-dominant bilinguals lack sensitivity to certain difficult Catalan-specific speech contrasts as previously reported by Pallier et al. (1997), but word pairs that are distinguished minimally by these nonnative contrasts share homophonous lexical representations.

A number of recent studies, however, have challenged this traditional view, reporting that L2 learners may establish a lexical contrast, even when they do not yet reliably discriminate the relevant phones (Cutler et al., 2006; Darcy et al., 2012; Escudero et al., 2008; Weber and Cutler, 2004). For example, Weber and Cutler (2004) used an eyetracking paradigm to investigate the mapping of phonetic information to lexical representations in Dutch learners of English. Weber and Cutler found an asymmetrical pattern of activation in which minimally contrasting syllables with /æ/ activated syllables with /ɛ/, but not vice versa (i.e. /pæn/ in target *panda* inappropriately activated *pencil*, but /pɛn/ in target *pencil* did not activate *panda*). The authors argue that this asymmetric pattern of lexical activation demonstrates that proficient Dutch learners of English indeed encode the /æ/–/ɛ/ contrast lexically; despite the difficulty they experience processing the contrast on perceptual tasks. They further hypothesized that orthographic information or metalinguistic knowledge may support lexical contrasts even when categorization of the contrast is not yet robust.

Escudero et al. (2008) found support for the proposed effect of orthography. They trained two groups of Dutch listeners on an artificial lexicon of novel English words containing /æ/ and /ɛ/. One group saw pictures of novel objects paired with the auditory form. For the other group, a written form of the novel word also accompanied the pictures and auditory forms. At test, listeners had to click on the picture that matched the auditory form that they heard. The eyetracking data revealed the same asymmetric pattern that has been reported by Weber and Cutler (2004), but crucially only for the group trained with both auditory and orthographic forms. The authors suggest that only the group trained with written forms was able to establish separate lexical representations for minimal pairs differing in the vowels /æ/ and /ɛ/.

Darcy et al. (2012) contributed to the debate by investigating the degree to which a purely grammatical explanation is supported. The authors investigated the perceptual categorization and lexical representation of the French vowel contrasts /y/–/u/ and /œ/–/ɔ/ by intermediate and advanced English learners. They found that while both learner groups differed from the native French speakers on the phonetic categorization of the vowel contrasts of interest, only the intermediate group demonstrated spurious priming for /y/–/u/ minimal pairs (suggesting that /y/–/u/ are not yet distinguished in the learner's phonological representations). Interestingly, while advanced learners appear to encode both contrasts in the phonological representations of lexical items, this ability coexists with suboptimal phonetic categorization performance for both French contrasts, challenging the common assumption that there is a one-to-one mapping between perceptual categorization and phonological representation. To account

Table 1. Feature specifications for the five Spanish vowels.

	i	e	a	o	u	(æ)
high	+	–	–	–	+	(–)
low	–	–	+	–	–	(+)
back	–	–	+	+	+	(–)

for these observed patterns, the authors propose the Direct Mapping from Acoustics to Phonology (DMAP) model of phonological acquisition. According to the DMAP, phonological acquisition requires only that acoustic correlates of phonological features be detected in the raw percepts, so as to trigger revisions to phonological feature matrices. The attunement of phonetic categories to the L2 input may occur after the phonological acquisition of contrasts.

The present study contributes to the debate regarding the nature of similarity in L2 phonology, by investigating whether phonological features are recombined to represent nonnative contrasts in the second language (Brown, 1998, 2000), and to what extent new phonological structure might be acquired. To this end, we examine the phonetic perception and lexical representation of the English-specific contrasts /i/–/ɪ/ and /ɑ/–/æ/ by advanced adult Spanish learners of English.

The Spanish vowel inventory contains five vowel phonemes /i, e, a, o, u/, whereas the English vowel inventory is larger with the precise number of vowel phonemes varying by dialect. Here we focus on two vowel contrasts that exist in English, but not in Spanish, namely tense/lax contrast /i/–/ɪ/ and front/back contrast /ɑ/–/æ/. These two contrasts are of interest to us here both because there are reports in the literature that suggest that they are notoriously problematic for Spanish learners of English in both perception and production, and because the contrastive feature specification of Spanish makes these an ideal test of Brown's feature-based model of L2 phonological acquisition.

According to Dresher (2009), contrastive feature specifications for a given language should be determined by a Successive Division Algorithm (which starts from an undifferentiated space and proceeds through successive divisions until all contrastive sounds in a language are distinguished in a given inventory). With respect to the Spanish vowel inventory this algorithm allows for two different analyses of Spanish vowels, differing in the scope of the [back] feature. Under one analysis, in which vowels are initially specified as [high] and [low], followed by [back], the [back] feature would not have scope over low vowels. That is, /a/ in this analysis would be neither front nor back.

An alternative analysis, however, would first divide the Spanish vowel inventory into front and back vowels. Successive division of the space would result in the feature specifications for Spanish vowels shown in Table 1. This analysis is also presented in Núñez Cedeño and Morales-Front (1999). Phonological evidence from the process of velar insertion in Spanish suggests that, as predicted by the latter, but not the former analysis of the Spanish vowel inventory, the vowel [a] in Spanish behaves as a member of the natural class of back vowels (d'Introno et al., 1988). We adopt this second analysis of the Spanish vowel system. This analysis of Spanish vowels also opens up the intriguing possibility that Spanish learners of English can capitalize on the implicit 6th category with the characteristics [–high, +low, –back].[2]

In sum, although the Spanish vowel inventory has segments similar to English /ɑ/ and /i/, it lacks the phonemes /æ/ and /ɪ/. Given that the feature that distinguishes the /ɑ/–/æ/, but not the /i/–/ɪ/ contrast, is present in the L1 feature inventory of Spanish speakers, it is interesting to ask whether the highly proficient L1 Spanish late-learners of English have acquired either or both of these nonnative contrasts. Brown's model predicts that the L1 feature [back] may be reused to represent the nonnative /ɑ/–/æ/ contrast, but that the /i/–/ɪ/ contrast will not be accurately perceived since the feature needed to make this tense/lax distinction is absent from the learner's L1 grammar (as are other potential contrasts which could be employed for this purpose, such as vowel length). Thus, Brown's model predicts that the /i/–/ɪ/ contrast will be unacquirable by Spanish learners, regardless of their experience.

A number of studies have investigated the perception of the vowel contrasts of interest by Spanish listeners, as well as speakers of other L1s that lack them. Flege (1991) had both experienced and inexperienced listeners perform an interlingual identification task in which listeners were asked to identify multiple tokens of /i/, /ɪ/, /ɛ/, and /æ/ spoken by 10 native speakers of American English with one of their five native language vowel categories, or to indicate that the vowel token did not sound like a Spanish vowel by selecting 'none'. The authors found that both English /i/ and /ɪ/ were regularly identified with Spanish /i/ and English /æ/ was typically identified with the Spanish vowel /a/. These results suggest that both of the sounds in these contrasts of interest would initially be assimilated to a single Spanish category and would be difficult for Spanish listeners to perceive and acquire. Escudero and Vasiliev (2011) also report that Spanish listeners assimilate Canadian English /æ/ to Spanish /a/.

A number of subsequent studies involving the English tense/lax contrast, including /i/–/ɪ/, have investigated which acoustic dimensions nonnative listeners attend to in perception. American English listeners are known to rely primarily on spectral properties, while duration plays only a secondary role (Hillenbrand et al., 2000). Fox et al. (1995) used multidimensional scaling (MDS) to investigate which phonetic properties Spanish and English listeners use to distinguish vowels. As their MDS model identified vowel height, but not duration, as a crucial dimension for Spanish listeners, the authors concluded that Spanish listeners do not have access to durational cues when perceiving vowels. They suggested that this could account for the difficulty that Spanish listeners encounter with /i/–/ɪ/. In contrast, later studies found that Spanish listeners do use vowel duration to discriminate English vowels (Bohn, 1995; Flege et al., 1997), though Flege et al.,1997 found that Spanish listeners rely on spectral cues as well. Interestingly, reliance on duration to distinguish English tense/lax vowels has also been reported for listeners of other L1s (German, Korean, Catalan, Mandarin, Japanese), many of which, like Spanish, lack relevant phonetic and phonological experience with length contrasts (Cebrian, 2006; Flege et al., 1997). Other more recent studies have demonstrated that Spanish listeners learn to use duration through exposure to the English language (Escudero and Boersma, 2004; Kondaurova and Francis, 2008; Morrison, 2008). These findings have led to the conclusion that durational cue use is developmental in Spanish listeners. Escudero (2006) proposed that Spanish listeners pass through four developmental stages on their way from no contrast to qualitatively English-like cue use.

It is important to point out that, to date, data regarding the development of Spanish speakers' phonetic and phonological acquisition of these contrasts have come from perceptual tasks, such as force-choice identification, discrimination and AXB. Importantly, as noted above, learners' ability to categorize and to lexically encode novel contrasts does not always go hand-in-hand (Darcy et al., 2012). Given our interest in the phonological representations of our Spanish learners, we employ a lexical decision task with repetition priming that requires that listeners access lexical items. In this way, the present study also brings new data to bear on the relationship between phonetic category acquisition and the acquisition of new lexical contrasts (i.e. English /i/–/ɪ/ and /ɑ/–/æ/). Two experiments were conducted using the same stimuli to explore the phonetic perception and lexical representation of the nonnative vowel contrasts by advanced L1 Spanish late-learners of English.

III Experiment 1

In Experiment 1, we used an AX discrimination task to investigate the ability of advanced Spanish late-learners of English to discriminate minimally contrastive English words and nonwords. We manipulated whether the experimental pairs were comprised of a vowel contrast that is common to both Spanish and English (i.e. /o/–/u/) or an English-specific vowel contrast (i.e. /ɑ/–/æ/ or /i/–/ɪ/). It was hypothesized that if feature availability predicts the perception and acquisition of nonnative contrasts as Brown's FBM suggests, we would find better performance for /ɑ/–/æ/ relative to /i/–/ɪ/.

1 Participants

Twenty-eight native English speakers (15 Male and 13 Female, Mean age = 21.2 years) and 28 advanced Spanish late-learners of English (11 Male and 17 Female, Mean age = 27.4 years) from the University of Maryland, College Park campus, USA, participated in this experiment. Native English-speaking participants were recruited from two introductory linguistics courses and were given course credit for their participation. The Spanish–English bilinguals were all students, post-docs or professors from various Spanish-speaking countries who were studying and/or working at the University of Maryland at the time of testing. All reported using English regularly throughout their day. On average the Spanish–English bilinguals who participated in this study began learning English in middle school at the age of 12, and they typically received 8 years of formal training. The mean length of residence in the USA was 6.5 years. All participants had very limited exposure to English in natural settings before arriving in the USA. All Spanish-speaking participants were compensated US$10 for taking part in this study.

2 Stimuli

The stimuli used in this experiment were monosyllabic English word and nonword pairs[3] that contained one of the following three vowel contrasts: /o/–/u/, /ɑ/–/æ/, and /i/–/ɪ/.

Sixteen minimal pairs were constructed for each contrast for a total of 96 test pairs. The full list of experimental items can be found in Appendix 1. All stimuli were recorded by one male and one female native speaker of American English. The condition in which each item appeared was counterbalanced across four lists such that a list contained either 'beat'–'beat', 'beat'–'bit', 'bit'–'beat', or 'bit'–'bit'.

3 Procedure

Following Brown (1998), we used an AX discrimination task to investigate the phonetic perception abilities of our L2 learners with respect to the three contrasts of interest. In order to make it difficult for our listeners to make their same-different judgments purely on the basis of low-level acoustic properties of the pair, AX pairs were comprised of two different voices. All participants heard stimulus A produced by a female speaker of American English, followed by X produced by a male speaker of the same variety. Moreover, our interstimulus interval between A and X was 1,500 ms, which should be long enough to favor phonemic processing (Werker and Logan, 1985).

Participants were assigned to one of the four lists according to their order of arrival and were tested individually in a sound-proof room seated in front of a Dell Inspiron 600 m laptop computer. DMDX was used to present stimuli and record each participant's response (Forster and Forster, 2003). At the beginning of each trial a fixation cross '+' appeared in the center of the screen to warn the participant that a stimulus was about to be presented. Next, the participant heard two auditory stimuli separated by a lag of 1,500 ms. For each pair, the participant was asked to decide, as quickly and accurately as possible, whether or not the stimuli he or she heard were the same or different. If A and X were the same, the participant was asked to respond by pressing the F key, if A and X were different, the participant was instructed to press the J key. Participants were allowed 4,000 ms to make their response. Response time was measured from the onset of the second auditory stimulus. Six practice items preceded the test trials to ensure that all participants had understood the instructions and were comfortable with the task. No feedback was given on either the practice or test trials and the data from these practice trials was not included in the analysis. The AX discrimination task was comprised of two 5-minute blocks (i.e. a block of English words, followed by a block of English non-words) and lasted approximately 10 minutes total.

4 Results

The data from one native English-speaking participant (S004) was excluded from analysis due to noncompliance with the experimental task (i.e. failing to respond to 37/48 trials in the second experimental block). Additionally, data from one Spanish learner of English (S115) was also excluded for failing to reach the criterion of a mean proportion accuracy of greater than .75 on the /o/–/u/ control contrast. Observations with response times (RTs) below 350 ms or for which RTs exceeding the allotted time (4,000 ms) were also excluded. Timeout responses made up 1.4% of the entire dataset. Response Accuracy (coded as '0 = incorrect' and '1 = correct' for each trial) was subsequently analysed separately for our control contrast /o–u/ and English-specific contrasts /ɑ/–/æ/ and /i/–/ɪ/

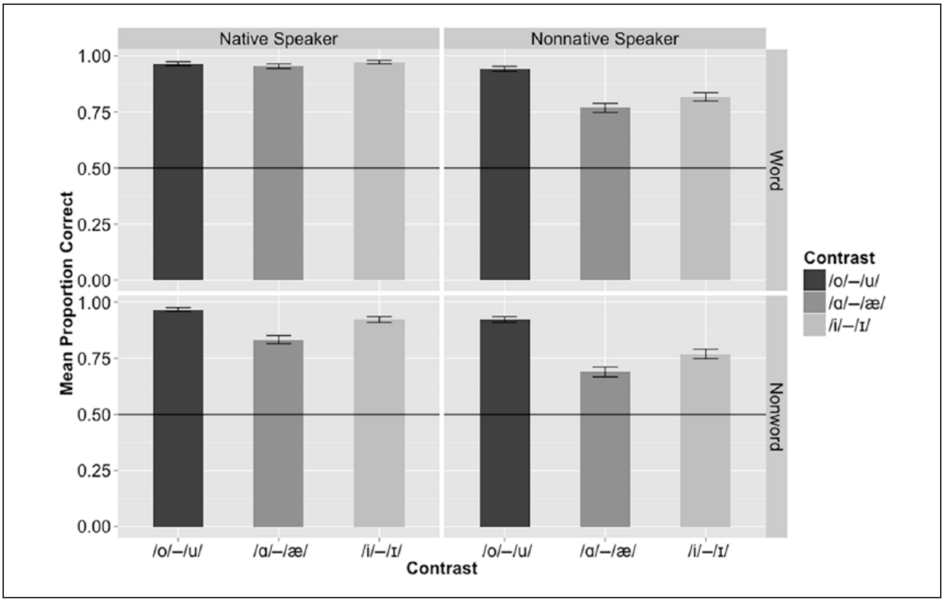

Figure 2. Mean proportion correct for word stimuli (top) and nonword stimuli (bottom) as a function of Language group (Native (left) vs. Nonnative Speaker (right)), and Contrast (/o/–/u/, /ɑ/–/æ/, and /i/–/ɪ/).
Note. Error bars show one standard error of the mean.

using a Generalized Linear Mixed Effects Model (GLMM) with Subject and Item as random effects. Figure 2 show mean proportion correct for word and nonword stimuli.

Statistical analyses were performed using GLMM binomial regression using R package lme4 (Bates et al., 2015) to assess the reliability of the effect of the experimental manipulation on Response Accuracy. Our analysis of response accuracy for the control contrast was comprised of fixed effects Language Group (native speaker vs. nonnative speaker), Lexical Status (word vs. nonword), Trial Type (same vs. different) and their interactions as fixed effects and Subject and Item as a random effect. For the control condition we found a main effect of Trial Type ($\beta = -2.7425$, SE = 1.0423, z-value = –2.631, $p = 0.00851$), such that participants were more accurate on different trials. Crucially, there was no main effect of Language Group ($\beta = -0.6956$, SE = 1.2336, z-value = –0.564, $p = 0.57286$), nor did Lexical Status ($\beta = -1.5021$, SE = 1.1485, z-value = –1.308, $p = 0.19093$) or any of the interaction terms reach significance. This analysis confirms the effectiveness of our control condition for which no effect of Language Group was predicted. Thus, our subsequent analyses were restricted to the English-specific contrasts /i/–/ɪ/ and /ɑ/–/æ/.

Analyses of response accuracy for the English-specific contrasts were comprised of fixed effects Language group (native speaker vs. nonnative speaker), Contrast (/i/–/ɪ/ vs. /ɑ/–/æ/), Lexical Status (word vs. nonword), Trial Type (same vs. different), their interaction terms, and Subject and Item as a random effect. We found a significant effect of

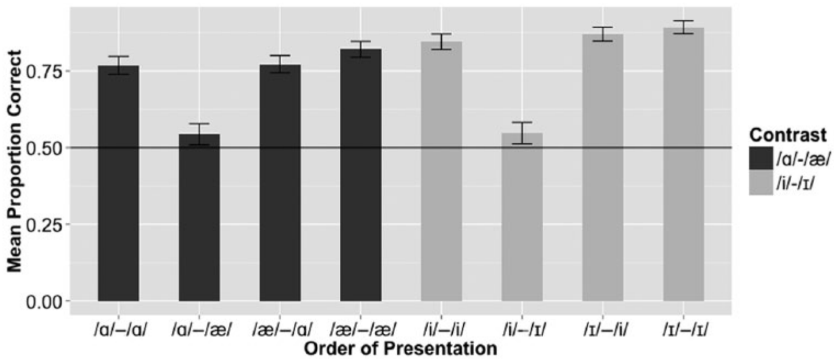

Figure 3. Mean proportion correct for the Nonnative speaker participants each of the four orders for the English-specific contrasts (/ɑ/–/æ/ and /i/–/ɪ/) averaged across word and nonwords stimuli.
Note. Error bars show one standard error of the mean.

Language Group (β = –1.89682, SE = 0.40255, z-value = –4.712, p = 2.45e-06). As expected, nonnative speakers were less accurate than native speakers on English-specific contrasts. The effect of Contrast was also significant (β = 1.36010, SE = 0.65993, z-value = 2.061, p = 0.0393). However, the effect was not in the predicted direction, as /ɑ/–/æ/ was found to be less accurate than /i/–/ɪ/. Also as expected, words were more accurate than nonwords, (β = –1.80566, SE = 0.37824, z-value = –4.774, p = 1.81e-06). There was also a significant Language Group × Lexical Status interaction (β = 0.84227, SE = 0.41473, z-value = 2.031, p = 0.0423), due to native speakers being less accurate on nonword stimuli. The three-way Contrast × Lexical Status × Trial Type interaction was also significant (β = 1.92422, SE = 0.95124, z-value = 2.023, p = 0.0431). None of the other interactions reached significance.

Two additional post hoc comparisons were made to consider whether the discrimination abilities of the Spanish learner group were susceptible to asymmetries depending on the order of presentation. Asymmetrical performance may be expected if learners are susceptible to perceptual magnet effects (Kuhl, et al. 1992), such that they have greater difficulty discriminating pairs when they hear a more prototypical vowel followed by a less prototypical one. Figure 3 shows the mean proportion correct for each of the 8 conditions (2 English-specific contrasts × 4 Presentation orders) of interest. The first post-hoc comparison tested whether the /ɑ/–/æ/ order differed from the other three possible orders of the /ɑ/–/æ/ contrast (i.e. /ɑ/–/ɑ/, /æ/–/ɑ/ and /æ/–/æ/). This pattern received statistical support (β = 3.7342, SE = .6675, z-value = 5.594, $p < .001$). A similar comparison was made for the /i/–/ɪ/ contrast. Of interest was whether the /i/–/ɪ/ order differed from the other three orders of the /i/–/ɪ/ contrast (i.e. /i/–/i/, /ɪ /–/i/ and /ɪ /–/ɪ /). Again, the order in which the more prototypical vowel [i] was followed by a less prototypical exemplar [ɪ] resulted in poorer performance than the two same pairings and the opposite order (β = 5.5470, SE = .7021, z-value = 7.900, $p < .001$). This suggests that learners' perception of the English-specific contrasts of interest is not symmetrical, but instead may be dependent on the order of presentation. In interpreting these findings, one should keep in mind that the present set of experiments were not designed to test for these asymmetries, nor were any explicit hypotheses made prior to the experiment. The results presented from this subset of data, while intriguing, are tentative at best and subsequent experimentation is necessary to test the perceptual magnet hypothesis directly.

5 Discussion

In Experiment 1, we tested the perceptual sensitivity of a group of advanced L1 Spanish late-learners of English with the expectation that the group's performance would differ on the English-specific contrasts as a function of feature availability. This prediction was based on a strong interpretation of Brown's hypothesis that states that, when available, features manipulated by the native grammar may be redeployed in the L2 to represent novel target language contrasts. If indeed this is the case, we could expect to observe superior performance (i.e. more accurate responses, fewer errors) for the /ɑ/–/æ/ contrast relative to the /i/–/ɪ/ contrast. We found no empirical support for Brown's predictions.

In general the results accord with previous findings. We found a reliable difference between native speakers and nonnative speakers for the English-specific contrasts /i/–/ɪ/ and /ɑ/–/æ/, but not the control contrast /o/–/u/. This result was expected given the large body of literature reporting the difficulty that nonnative speakers experience with certain nonnative contrasts.

Unexpectedly, however, we found that the Spanish–English bilinguals did not differ significantly in their performance on the /ɑ/–/æ/ contrast relative to the /i/–/ɪ/ contrast. While this was an unexpected result, there is a possibility that this effect may have been driven by factors that were external to the experiment, namely, the fact that there are a number of English words, such as the word 'pasta', in which the initial vowel may be pronounced as either [ɑ] and [æ] as a function of dialect. Alternations of this sort have been shown to reduce the contrastiveness of otherwise distinctive sound pairs (Huang and Johnson, 2011; Johnson and Babel, 2010). We speculate that this may have contributed to the decrement in accuracy observed for the /ɑ/–/æ/ for nonwords in the native speaker group. While the exact nature of the effect observed for /ɑ/–/æ/will have to be taken up in future investigation, it is clear that one should be cautious in interpreting the relative accuracy of the two English-specific contrasts for the nonnative speakers given that lower accuracy was also observed for /ɑ/–/æ/ for the native speaker group for whom the contrast was not expected to be problematic. Note that this finding itself suggests that there is more to similarity than acoustic or feature-based similarity alone. This finding suggests that perceived similarity must also take the contrastiveness of a pair of sounds into account, as Johnson and Babel (2010) also suggest.

IV Experiment 2

Experiment 2 used a medium-term repetition priming paradigm (Pallier et al., 2001) to investigate whether advanced Spanish learners of English show evidence of having acquired distinct phonological representations for minimal pairs in their L2 lexicon. In this task, participants are asked to perform an auditory lexical decision task on one of four counterbalanced lists of English words and nonwords. Test words and nonwords were followed 9 to 23 items later by either an identical stimulus (i.e. Repetition, such as ‘sheep’–‘sheep’) or a minimal pair (i.e. Minimal Pair, such as ‘sheep’–‘ship’). Each stimulus pair was distinguished minimally by either a common vowel contrast (i.e. /o/–/u/) or an English-specific vowel contrast (i.e. /ɑ/–/æ/ or /i/–/ɪ/). It was predicted that if Spanish-speaking participants have acquired distinct lexical representations for test pairs containing minimal contrasts, then facilitation effects should be observed for the Repetition condition, but not for the Minimal Pair condition. However, if minimal pairs containing nonnative contrasts have homophonous lexical representations, facilitation effects of approximately the same magnitude should be observed for nonnative contrasts in both conditions. Crucially, if feature availability predicts the acquisition of nonnative contrasts, as Brown’s FBM suggests, we should find minimal pair priming for /i/–/ɪ/, but not /ɑ/–/æ/.

1 Participants

Participants in this experiment were the same 28 Spanish speakers and 28 native English speakers who participated in the AX discrimination task described for Experiment I. All participants performed the auditory lexical decision task prior to performing the AX discrimination task.

2 Stimuli

The stimuli used in the medium-term repetition priming experiment were the same 96 experimental word and nonword pairs used in the auditory AX discrimination task detailed above. In addition to the 192 test items, 32 additional words and 32 nonwords served as fillers. Crucially, filler items did not include any of the vowel contrasts of interest. The condition (Minimal pair or Repetition) in which each item appeared was counterbalanced across four lists such that a list contained either ‘beat’–‘beat’, ‘beat’–‘bit’, ‘bit’–‘beat’, or ‘bit’–‘bit’. Four counterbalanced lists of 256 stimuli were created. Each list contained one member of each stimulus pair followed, 9 to 23 items down the list, by either a repetition or a minimal pair, spoken by a different speaker.

Every attempt was made to match word frequency across the word stimuli for the three contrasts. Given other constraints on stimulus selection, such as the need to find common monosyllabic minimal pairs contrasting the sounds of interest, it was not possible to match these absolutely. We considered two spoken word frequency measures from the SUBTLEXus word frequency database (Brysbaert and New, 2009), SUBTLwf and Lg10wf. SUBTLwf is the word frequency counts per million words, whereas Lg10wf is a Log10 transformed frequency count. Table 2 provides a summary of the frequency distributions for the contrasts used in the minimal pairs. Two separate one-way ANOVAs compared SUBTLwf and Lg10wf for the three contrasts. Both failed to reach significance ($F(2,93) = 0.592$, $p = 0.555$; $F(2,93) = 1.066$, $p = 0.349$).

It is also worth mentioning that it is unclear whether and how these frequency measures relate to L2 learners’ familiarity with the words. (This important point is also made

Table 2. Summary of word frequency distributions by Contrast.

Contrast	SUBTLwf (SE)	Lg10wf (SE)
/o/–/u/	62.5 (18.18)	3.08 (.110)
/ɑ/–/æ/	204.9 (162.86)	3.15 (.118)
/i/–/ɪ/	100.26 (26.91)	3.31 (.108)

by Darcy et al., 2012.) For this reason, learners were also asked to indicate their (un-) familiarity with the test words at the end of the experiment. Learners were given a list of the test stimuli and were asked to circle any unknown words. As a group, learners' reported 98.5% of words to be familiar.

3 Procedure

Participants were assigned to one of the four lists according to their order of arrival and were tested individually in a sound-proof room seated in front of a Dell Inspiron 600 m laptop computer. DMDX (Forster and Forster, 2003) was used to present stimuli and record participants' responses. Participants responded by pressing one of two buttons on the keyboard. At the beginning of each trial a fixation cross '+' appeared in the center of the screen to warn the participant that a stimulus was about to be presented. Next, participants heard an auditory stimulus. For each stimulus, the participant was asked to decide, as quickly and accurately as possible, whether or not the stimulus they heard was an English word. If the stimulus was a word, the participant was asked to respond by pressing the F key, if the sequence was not an English word, the participant was instructed to press the J key. Following Pallier et al. (2001), stimuli were presented with an interstimulus interval of 2.5 seconds. Response time was measured from the onset of each auditory stimulus. The test trials were preceded by six practice items to ensure that the participant understood the experimental task. No feedback was provided on either the practice or test items. The auditory lexical decision task lasted approximately 20–25 minutes and was broken into two 10–12 minute blocks separated by a self-timed break.

4 Results

The data from two Spanish learners of English (S119 and S125) was excluded from the analysis for failing to reach our criterion of a d' score of 1 for the /o/–/u/ control contrast. Data points which fell within the predetermined low and high cutoff of 300 ms and 2,500 ms were retained for an analysis of 'Repetition effect' (86.2% of the original data). Following Pallier et al., we define a repetition effect as a reaction time decrease between the first and second occurrence of an item or between the occurrence of an item and its minimal pair. Repetition effects were computed by subtracting the RT to the second occurrence of an item (i.e. 'Target') from the initial occurrence (i.e. 'Prime'), which served as its baseline. Thus, a positive value indicated a priming effect. Figure 4 shows repetition effects for word stimuli.

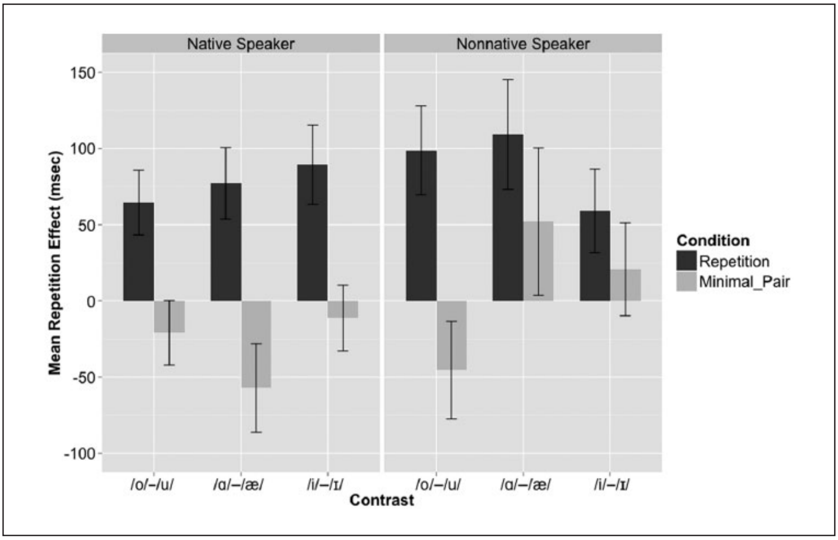

Figure 4. Mean repetition effects in response to word stimuli as a function of Language group (Native (left) vs. Nonnative Speaker (right)), Condition (Repetition vs. Minimal Pair), and Contrast (/o/–/u/, /ɑ/–/æ/, and /i/–/ɪ/).
Note. Error bars show one standard error of the mean.

Linear mixed effects modeling was employed to assess the reliability of the effects elicited by the experimental manipulations (Baayen, 2008). Analyses of repetition effects for word stimuli for the control contrast /o/–/u/ comprised of fixed effects Language Group (native speaker vs. nonnative speaker), Condition (repetition vs. minimal pair), as well as the Language Group × Condition interaction and Subject and Item as random effects. A main effect of Condition ($\beta = 85.23$, $SE = 34.01$, t-value = 2.506, $p = 0.0125$) was observed. As expected, repetition effects were larger for the repetition condition than for the minimal pair condition. Additionally, there was no effect of Language Group ($\beta = -24.63$, $SE = 38.11$, t-value = –0.646, $p = 0.5191$), nor did the interaction reach significance, suggesting that there was no difference between the native and nonnative group with respect to their performance on the control contrast /o/–/u/. This confirms the adequacy of the control condition for which no differences were expected.

Analyses of repetition effects for word stimuli for the English-specific contrasts were again comprised of fixed effects Language Group (native speaker vs. nonnative speaker), Condition (repetition vs. minimal pair), Contrast (/ɑ/–/æ/ vs. /i/–/ɪ/), as well as all two-way and three-way interactions and Subject and Item as random effects. A main effect of Condition ($\beta = 134.58$, $SE = 40.20$, t-value = 3.348, $p = 0.000837$) was again observed, with Repetition yielding more priming than Minimal Pairs, as expected. We also observed a main effect of Language Group ($\beta = 110.04$, $SE = 45.41$, t-value = 2.424, $p = 0.015497$), with nonnative speakers showing more priming overall. No other significant effects were found. Statistical analyses of nonword stimuli revealed no significant results for either the control or English-specific contrasts regardless of Language group, Contrast or Condition.

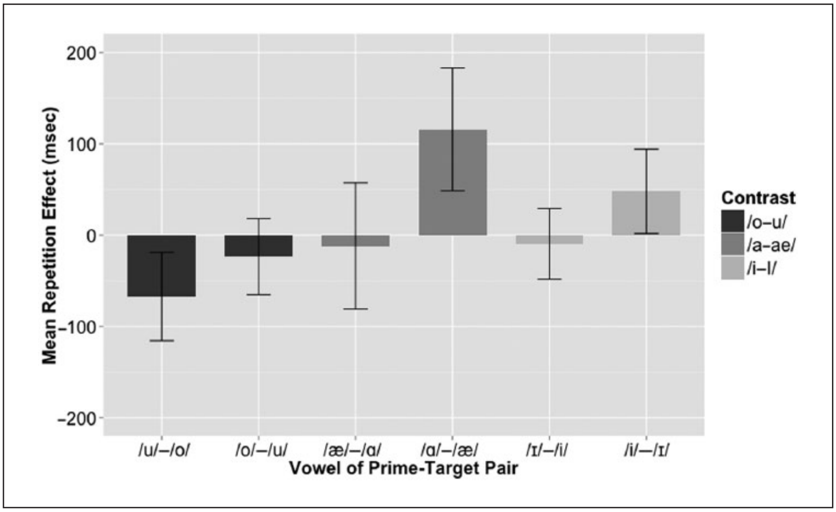

Figure 5. Mean repetition effects in response to word stimuli for the Nonnative Speaker group for the Minimal Pair condition.
Note. Error bars show one standard error of the mean.

In order to further assess the possible confounding influence of word frequency on Priming, two additional linear mixed-effects models were fit to the repetition priming data for the experimental contrasts of interest. Both of these models included fixed effects Language Group, Condition, and Contrast, as well as their interaction terms; one of them included Lg10wf as an additional fixed effect; another included Lg10wf as a fixed effect as well as all possible interaction terms with the other fixed effects in the model. A likelihood ratio test comparing our original model without a fixed effect for Lg10wf to the more complex model with an additional parameter for Lg10wf revealed that the more complex model did not provide a significantly better fit of the data than the simpler model ($\chi^2(1) = .1932$, $p = .6603$), and thus the inclusion of Lg10wf in the model is not statistically justified. A second likelihood ratio test revealed that the more complex model including Lg10wf as main effect as well as all possible interaction terms with the other fixed effects in the model likewise did not provide a significantly better fit of the data than the simpler model ($\chi^2(8) = 11.788$, $p = .1609$). It should be noted that both analyses use a null hypothesis significance testing approach (and so the null hypothesis cannot be confirmed); however, the results from both analyses are consistent in showing no confounding influence of frequency in our data.

Finally, as was done for the discrimination data presented above, we also considered whether there were asymmetries in the amount of priming observed for the Minimal Pair Condition for the Nonnative Speakers. The priming results for each of the three contrasts for the Minimal Pair Condition are summarized in Figure 5. Of particular interest are the English-specific contrasts where a previous encounter with a Prime could result in differential amount of priming in the Target as a result of order of presentation. That is, it may be expected that the Prime ‘beat’ could result in facilitated processing of the minimal pair Target 'bit', but not vice versa. It appears that minimal pair priming was only observed in our nonnative listeners when targets containing /ɪ/ were preceded by primes containing /i/, and when targets containing /æ/ were preceded by primes containing /ɑ/, but not vice versa.

5 Discussion

Experiment 2 investigated whether advanced L1 Spanish late-learners of English have acquired distinct lexical representations for English minimal pairs which are distinguished by the English-specific vowel contrasts /ɑ/–/æ/ and /i/–/ɪ/ using a medium-term repetition priming paradigm. We hypothesized that if learners are unable to acquire new features, but are able to redeploy phonological features from their L1, the advanced Spanish late-learners of English should differ in their performance on these two nonnative contrasts. Given that the Spanish learners' native grammar makes available the feature [back], but the language lacks a tense/lax distinction, the learners were expected to perform better (i.e. show no/less minimal pair priming) for the /ɑ/–/æ/ contrast relative to the /i/–/ɪ/ contrast. This prediction was not borne out.

For word and nonword stimulus pairs that contain a native contrast (i.e. /o/–/u/), we replicate previous findings. Like Pallier et al. (2001), we observe larger repetition effects for the repetition condition than the minimal pair condition for words containing contrasts that are common to both languages, but no reliable repetition effects for nonword repetitions or minimal pairs. Moreover, no difference in priming was observed between Spanish participants and native English speaker controls for the control condition.

The results for words containing nonnative contrasts are less clear. Recall that it was hypothesized that the availability of a [back] feature in the Spanish grammar may aid learners in establishing novel segmental representations so as to distinguish /ɑ/–/æ/. Therefore, less minimal pair priming was expected for /ɑ/–/æ/ than for /i/–/ɪ/. Although we found a main effect of Condition and of Language Group, we found no significant interaction effects. Thus, while visual inspection of the data may lead one to believe that priming is found for the Spanish participants in the minimal pair condition for the /ɑ/–/æ/ contrast, but not the /i/–/ɪ/ contrast, we have little statistical support for this claim. Moreover, note that such results would also run counter to the predictions of Brown's model regarding the acquirability of nonnative contrasts. Our repetition priming results for nonnative contrasts are at odds with previous findings from Pallier et al. (2001), who report significant minimal pair priming for Spanish-dominant Spanish–Catalan bilinguals for Catalan-specific contrasts. Both of these points will be taken up in greater detail in the section that follows.

V General discussion

Two experiments were conducted to investigate the phonetic perception and lexical representation of two English-specific vowel contrasts (i.e. /ɑ/–/æ/ and /i/–/ɪ/) by advanced Spanish late-learners of English. These experiments were designed to explore the hypothesis that the perception and acquisition of novel lexical representations of nonnative contrasts is contingent upon the availability of phonological features in a learner's native grammar. In Experiment 1, an AX discrimination task was employed to investigate the Spanish participants' ability to discriminate minimally contrastive English words and nonwords. In Experiment 2, we used medium-term repetition priming in an auditory lexical decision task to investigate word-recognition processes in the same participants and using the same stimuli. In both experiments it was expected that, if feature availability predicts the acquirability of nonnative contrasts by L2 learners, we should observe better performance (i.e. more accurate responses in Experiment 1 and less priming in Experiment 2) for minimal pairs containing the /ɑ/–/æ/ contrast relative to the /i/–/ɪ/ contrast. In both experiments we found converging evidence that (1) the Spanish learners of English who participated in our study continued to differ from native speakers of English in their sensitivity to the two English-specific vowel contrasts that we tested. This finding was expected, since differences between native and nonnative speakers are ubiquitous in the SLA literature. Yet, (2) these participants performed much better on both tasks than was expected based on past reports. On the AX discrimination task, the mean proportion correct for word stimuli for the Spanish participants was .82 and .77 for the /i/–/ɪ/ and /ɑ/–/æ/ contrasts, respectively. For nonwords, our participants scored .77 and .69. This is well above chance (.5) and much better than would be expected from participants who were experiencing extreme difficulty with these contrasts. Moreover, in contrast with Pallier et al. (2001), we did not observe statistically reliable minimal pair priming by our Spanish learners of English for words differing in a single English-specific contrast.

This second finding is surprising given reports that (1) early learners outperform late-learners in both perception and production tasks (Flege and MacKay, 2004; Piske et al., 2002), and (2) early Spanish-dominant Spanish–Catalan bilinguals, with extensive experience with Catalan, experience persistent difficulty with certain Catalan-specific contrasts. The finding that our highly proficient Spanish late-learners of English performed better than the Spanish-dominant Spanish-Catalan bilingual participants in Pallier et al. (2001) might be taken to suggest better language separability on the part of the Spanish–English as opposed to Spanish–Catalan bilinguals. More work is clearly needed in order to understand the cause of these differences.

Our findings also shed light on the nature of the phonological form of these learners' lexical representations. First, our results are incompatible with a story in which these learners systematically fail to perceive or to represent these novel contrasts in their L2 lexicons, as previously suggested to account for the Pallier et al. (2001) findings. That is, the Spanish participants in our study cannot simply be conflating the contrast between English /i/ and /ɪ/ or /ɑ/ and /æ/ in perception across the board, nor can they be representing minimal pairs such as /ʃip/ 'sheep' and /ʃɪp/ 'ship' as /ʃip/.

With respect to the role of distinctive features in L2 perception and phonological acquisition, Experiments 1 and 2 were designed to test the predictions of Brown's hypothesis that the phonological features employed by the native grammar may be redeployed in the L2 to represent novel contrasts. To investigate this, we took advantage of the fact that the native grammar of Spanish makes available the necessary features to represent /ɑ/–/æ/, but not /i/–/ɪ/. However, in both experiments, we failed to find the predicted difference between /i/–/ɪ/ than /ɑ/–/æ/. In fact, numerically speaking, the observed effects were in the opposite direction. The results of Experiments 1 and 2, therefore, provide converging evidence, suggesting that a strong version of Brown's hypothesis is likely to be incorrect. In particular, our results suggest that not only were some available features not redeployed as predicted, but that new features are indeed acquirable by late-learners (Curtin et al., 1998; González Poot, 2011; LaCharité and Prévost, 1999).

So, what kind of contrast was acquired? This question cannot be addressed with certainty here. On the one hand, it is possible that the learners in our study have established target-like phonological representations for both contrasts (i.e. a [±tense] contrast for /i/–/ɪ/ and a [±back] contrast for /ɑ/–/æ/). Our findings are also compatible with an alternative explanation. Learners may have constructed L2 representations that are distinct from those thought to be used by native speakers of the language, but which nevertheless serve to distinguish L2 lexical items. A plausible candidate is length, since Spanish listeners have been shown to rely on duration cues on perceptual tasks like those reviewed earlier. An interesting possibility is that learners may represent the contrast between /i/–/ɪ/ as one of length in which English [i] is represented as bimoraic and English [ɪ] as monomoraic. This suprasegmental solution is qualitatively different from the representation of feature-based contrasts in L2 phonology, but would nevertheless serve the learner, allowing them to represent the contrast in L2 words without having constructed native-like feature-based representations that require segmental features not employed in the listeners' native grammar. As suggested to us by an anonymous reviewer, a similar analysis presents itself for /ɑ/–/æ/, since the presence of monosyllabic words such as 'spa' and 'bra' and the absence of forms such as *[bræ] and *[spæ] have been taken as evidence that the former but not the latter meet the English minimal word requirement (i.e. bimoraicity). Thus, it is possible that Spanish listeners represent one or both of the contrasts as one of length. Mah and Archibald (2002) report that English native speakers are capable of encoding length contrasts in their representations of L2 Japanese geminates, even though English does not contrast segmental length phonologically (see also Hayes-Harb and Masuda, 2008). Future work should attempt to tease apart these possibilities by exploring individual learners' cue weighting preferences along side their ability to lexically encode the contrasts using methods such as repetition priming or eye-tracking, which have proven to be useful tools.

Turning now to the relationship between phonetic category learning and the acquisition of phonological contrast, while our participants appear to succeed in establishing phonological contrasts that serve to distinguish L2 words (either by establishing distinct segmental or suprasegmental representations), their performance continues to differ from native speakers of English on English-specific contrasts. We believe that these residual differences reflect the need for further attunement of phonetic categories to the L2 input. This type of asymmetry is expected in models of phonological acquisition, such as the DMAP (Darcy et al., 2012), in which the acquisition of phonological contrasts and the attunement of phonetic categories occur at two disjoint levels. Furthermore, our analyses of order of presentation suggest that there are asymmetries in the learner's phonetic processing of both contrasts. In particular, it was found that at the phonetic processing level, one of the categories in each pair is dominant, resulting in perceptual magnet like effects. This observed category dominance at the phonetic processing level may be determined by acoustic–phonetic proximity to the nearest L1 category (Cutler et al., 2006), or by metalinguistic influences such as orthographic information.[4] Future work should further investigate this dominance pattern to determine its causes in the Spanish speaker case.

Finally, our experiments were not designed to test whether discrimination patterns can be predicted via detailed acoustic analyses of the target language and native language sound categories. These questions must be left up to future investigation. However, a superficial comparison at least suggests that doing so would not make the correct predictions for our data presented here because Spanish /i/ is closer acoustically to English /ɪ/ than Spanish /ɑ/ is to English /æ/. This is perhaps not that surprising since the proposal is typically thought to hold for naive listeners or beginners at the initial stage of acquisition. The Spanish participants in our study are experienced learners who have had substantial exposure to the target language. As such, their responses are likely to be impacted by other levels of linguistic knowledge that they have acquired. As others have suggested, we argue that acoustic similarity is only one of several factors that contribute to perceived similarity in experienced listeners (Johnson and Babel, 2010). Future work should explore how the factors that influence perceptual similarity may change with proficiency level, as well as the type and quantity of training and exposure. In future work on L2 phonology it will be important to pin down the exact role played by acoustic similarity, acquired knowledge of phonological contrast, lexical knowledge, as well as a listeners' perceptual ability for a given contrast.

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Notes

1. Acoustic comparisons of vowels have traditionally been made between mean F1 and F2 values, which are often plotted for visualization in a two dimensional space (see, among others, Bradlow, 1995; Escudero and Chládková, 2010; Flege, 1991). Yet, it is important to point out that a range of other acoustic properties may also contribute to perceived similarity, including additional spectral information (F0, higher formants) or temporal information (duration) associated with each vowel production. Moreover, alternative scales, such as Mels and Barks, have been argued to be psychoacoustically more realistic units for such comparisons (Stevens et al., 1937; Zwicker, 1961). As discussed in Cebrian (2002), an important step for researchers who are performing acoustic comparisons is to identify the crucial cues and find a precise, consistent, and reliable way of measuring them (for further discussion of methodological issues associated with acoustic comparisons of vowel categories, see Strange, 2007).

2. Given that *[+high +low] is impossible, the implicit 3 × 2 feature system implies the (potential) existence of [–high +low –back].
3. One bisyllabic word pair was included for the /ɑ/–/æ/ contrast (i.e. 'bottle' - 'battle'). This exception was made for consistency in the number of items across the three experimental contrasts.
4. As an anonymous reviewer points out, the vowel /æ/ is written with <a> in English, which is how /a/ is written in Spanish. Likewise /ɪ/ is represented by the letter <i>, the exact letter used to represent Spanish /i/. Thus, orthographic factors may likewise hinder the formation of target-like phonetic categories representations.

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Appendix 1. Experimental stimuli used in Experiments 1 and 2.

	/o/	/u/	/ɑ/	/æ/	/i/	/ɪ/
1	blow	blue	odd	add	bean	bin
2	boat	boot	bond	band	beat	bit
3	coal	cool	bottle	battle	cheap	chip
4	crow	crew	block	black	deep	dip
5	float	flute	blond	bland	feel	fill
6	glow	glue	con	can	feet	fit
7	hope	hoop	cop	cap	jean	gin
8	known	noon	cot	cat	lead	lid
9	mode	mood	hot	hat	leak	lick
10	pole	pool	mop	map	leap	lip
11	road	rude	pot	pat	leave	live
12	role	rule	rock	rack	reach	rich
13	show	shoe	rot	rat	scene	sin
14	soap	soup	sock	sack	seek	sick
15	toll	tool	shock	shack	sheep	ship
16	wrote	root	top	tap	sleep	slip
	/o/	/u/	/ɑ/	/æ/	/i/	/ɪ/
1	botʃ	butʃ	bɑf	bæf	tʃim	tʃɪm
2	bof	buf	bɑv	bæv	div	dɪv
3	bok	buk	fɑm	fæm	gid	gɪd
4	fop	fup	fɑp	fæp	gip	gɪp

Appendix 1. (Continued)

	/o/	/u/	/ɑ/	/æ/	/i/	/ɪ/
5	gob	gub	gɑm	gæm	kib	kɪb
6	kog	kug	gɑn	gæn	kig	kɪg
7	log	lug	kɑg	kæg	mip	mɪp
8	lov	luv	mɑf	mæf	nib	nɪb
9	mof	muf	mɑz	mæz	pib	pɪb
10	moz	muz	nɑs	næs	riv	rɪv
11	nob	nub	nɑz	næz	sig	sɪg
12	nog	nug	pɑf	pæf	ʃig	ʃɪg
13	pon	pun	sɑn	sæn	tid	tɪd
14	pov	puv	ʃɑθ	ʃæθ	tig	tɪg
15	rok	ruk	tɑf	tæf	θig	θɪg
16	tof	tuf	zɑt	zæt	θip	θɪp